PHY101: Introduction to Physics-1

Monsoon, 2018

Tutorial-13

1. One cubic meter of atomic hydrogen at 0°C and atmospheric pressure contains approximately 2.7×10^{25} atoms. The first excited state of the hydrogen atom has an energy of 10.2 eV above the lowest energy level, called the ground state. Use the Boltzmann factor to find the number of atoms in the first excited state at 0°C and at 10000°C.

2. At what temperature would the average speed of helium atoms equal (a) the escape speed from Earth, 1.12×10^{4} m/s and (b) the escape speed from the Moon, 2.37×10^{3} m/s? Note that the mass of a helium atom is 6.64×10^{-27} kg.

Copy of quiz 2 will be shown in the tutorial class after the discussion of these two problems.